

REBECCA HWA

Department of Computer Science
School of Engineering and Applied Science
The George Washington University, DC
(202) 994-6779
rebecca.hwa@gwu.edu

EDUCATION

PhD in computer science, 2001, Harvard University (Advisor: Stuart Shieber).
BS in computer science and engineering (summa cum laude), 1993, UCLA.

PROFESSIONAL EXPERIENCE

Professor and Department Chair, Department of Computer Science, The George Washington University, DC (2023 – present)
Program Officer, Directorate for Computer and Information Science and Engineering, National Science Foundation, VA (2019 -- 2023)
Professor, Department of Computer Science, University of Pittsburgh, PA (2020 – 2023)
Associate Professor, Department of Computer Science, University of Pittsburgh, PA (2009 – 2020)
Assistant Professor, Department of Computer Science, University of Pittsburgh, PA (2003 – 2009)
Postdoctoral Fellow, Institute for Advanced Computer Studies, University of Maryland at College Park, MD (2001 – 2003)
Independent Contractor, Applied Language Technologies, Inc. (now a part of Nuance), Cambridge, MA (1995 – 1996).

HONORS

National Science Foundation CAREER Award (2008).
Department of Computer Science Teaching Award, University of Pittsburgh (2004, 2007, 2016, 2017).
National Science Foundation Graduate Fellowship (1993–1996).
Outstanding Undergraduate in Computer Science, UCLA (1993).

RESEARCH, SCHOLARLY AND CREATIVE ACTIVITIES

REFEREED JOURNAL ARTICLES

(J.1) Ian J. Barbash, Billie S. Davis, Melissa Saul, Rebecca Hwa, Emily B. Brant, Christopher W. Seymour, and Jeremy M. Kahn. 2024. Association between Medicare’s sepsis

- reporting policy (SEP-1) and the documentation of a sepsis diagnosis in the clinical record. *Medical care* 62, no. 6 (2024): 388-395.
- (J.2) Yongsu Ahn, Muheng Yan, Yu-Ru Lin, Wen-Ting Chung, and Rebecca Hwa. 2022. Tribe or Not? Critical Inspection of Group Differences Using TribalGram. *ACM Transactions on Interactive Intelligent Systems (TiiS)*, 12(1), 1-34.
 - (J.3) Omid Kashefi, Tazin Afrin, Meghan Dale, Christopher Olshefski, Amanda Godley, Diane Litman, Rebecca Hwa. 2022. ArgRewrite V. 2: An Annotated Argumentative Revisions Corpus. *Language Resources and Evaluation*, 1-35.
 - (J.4) Gaurav Trivedi, Phuong Pham, Wendy Chapman, Rebecca Hwa, Janyce Wiebe, and Harry Hochheiser. 2018. NLPReViz: An Interactive Tool for Natural Language Processing on Clinical Text. *Journal of the American Medical Informatics Association*. Vol. 25, No. 1.
 - (J.5) Joshua S. Albrecht, Rebecca Hwa, and G. Elisabeta Marai. 2009. The Chinese Room: Visualization and Interaction to Understand and Correct Ambiguous Machine Translation. *Computer Graphics Forum*, Vol. 28, No. 3.
 - (J.6) Joshua S. Albrecht and Rebecca Hwa. 2008. Regression for Sentence-Level Machine Translation Evaluation. *Machine Translation* Vol. 22, No. 1.
 - (J.7) Kailong Liu, Wendy Chapman, Rebecca Hwa, and Rebecca Crowley. 2007. Heuristic Sample Selection to Minimize Reference Standard Training Set for a Part-of-Speech Tagger. *Journal of the American Medical Informatics Association* Vol. 14, No. 5.
 - (J.8) Theresa Wilson , Janyce Wiebe, and Rebecca Hwa. 2006. Recognizing Strong and Weak Opinion Clauses. *Computational Intelligence* Vol. 22, No. 2.
 - (J.9) Rebecca Hwa, Philip Resnik, Amy Weinberg, Clara Cabezas, and Okan Kolak. 2005. Bootstrapping Parsers via Syntactic Projection across Parallel Texts. *Special Issue of the Journal of Natural Language Engineering on Parallel Texts*, Eds. Rada Mihalcea and Michel Simard. Vol. 11, No. 3.
 - (J.10) Rebecca Hwa. 2004. Sample Selection for Statistical Parsing. *Computational Linguistics*. Vol. 30, No. 3.
 - (J.11) Bonnie J. Dorr, N. Fazil Ayan, Nizar Habash, Nitin Madnani, and Rebecca Hwa, 2003. Rapid Porting of DUSTER to Hindi. In *ACM Transactions on Asian Language Information Processing (TALIP)*, Vol. 2, No. 3.

REFEREED CONFERENCE PAPERS

- (C.1) Rr. Nefriana, Muheng Yan, Rebecca Hwa, and Yu-Ru Lin. 2026. "Leader-driven or Leaderless: How Participation Structure Sustains Engagement and Shapes Narratives in Online Hate Communities," In *The Proceedings of the 20th International AAAI Conference on Web and Social Media (ICWSM-2026)*, Los Angeles, California.

- (C.2) Meiqi Guo, Rebecca Hwa, Adriana Kovashka. 2023. “Decoding Symbolism in Language Models,” In *The Proceedings of the Association for Computational Linguistics (ACL-2023)*, Toronto, Canada.
- (C.3) Mingda Zhang, Adriana Kovashka, Rebecca Hwa. 2023. “How to Practice VQA on a Resource-limited Target Domain”, In *The Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*.
- (C.4) Meiqi Guo, Rebecca Hwa, Adriana Kovashka. 2021. “Detecting Persuasive Atypicality by Modeling Contextual Compatibility,” In *The Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV-2021)*.
- (C.5) Mingda Zhang, Tristan Maidment, Ahmed Diab, Adriana Kovashka, Rebecca Hwa. 2021. “Domain-robust VQA with Diverse Datasets and Methods but No Target Labels”, In *The Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR-2021)*.
- (C.6) Tazin Afrin, Omid Kashefi, Christopher Olshefski, Diane Litman, Rebecca Hwa, and Amanda Godley. 2021. Effective Interfaces for Student-Driven Revision Sessions for Argumentative Writing. In *The Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI '21)*.
- (C.7) Carter Burns, Frederick L. Crabbe and Rebecca Hwa. 2021. Let’s Do the Time Warp Again: Human Action Assistance for Reinforcement Learning Agents. In *The Proceedings of the 13th International Conference on Agents and Artificial Intelligence - Volume 1: ICAART*, pages 92-100.
- (C.8) Meiqi Guo, Rebecca Hwa, Yu-Ru Lin and Wen-Ting Chung. 2020. Inflating Topic Relevance with Ideology: A Case Study of Political Ideology Bias in Social Topic Detection Models. In *The Proceedings of The 28th International Conference on Computational Linguistics (COLING-2020)*.
- (C.9) Muheng Yan, Yu-Ru Lin, Rebecca Hwa, Ali Mert Ertugrul, Meiqi Guo and Wen-Ting Chung. 2020. MimicProp: Learning to Incorporate Lexicon Knowledge into Distributed Word Representation for Social Media Analysis. In *The Proceedings of ICWSM-20 (the 14th International AAAI Conference on Web and Social Media)*.
- (C.10) Changsheng Liu and Rebecca Hwa. 2019. A Generalized Idiom Usage Recognition Model based on Semantic Compatibility. In *The Proceedings of the Thirty-Third AAAI Conference on Artificial Intelligence (AAAI 2019)*, Honolulu, Hawaii.
- (C.11) Mingda Zhang, Adriana Kovashka, and Rebecca Hwa. 2018. Equal But Not The Same: Understanding the Implicit Relationship Between Persuasive Images and Text. In *The Proceedings of the 29th British Machine Vision Conference (BMVC)*, Newcastle, UK.
- (C.12) Changsheng Liu and Rebecca Hwa. 2018. Heuristically informed unsupervised idiom usage recognition. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pp. 1723-1731. Austin, Texas.

- (C.13) Homa B. Hashemi and Rebecca Hwa. 2018. Jointly Parse and Fragment Ungrammatical Sentences. In *The Proceedings of Thirty-Second AAAI Conference on Artificial Intelligence*, New Orleans, Louisiana.
- (C.14) Fan Zhang, Homa B. Hashemi, Rebecca Hwa and Diane Litman. 2017. A Corpus of Annotated Revisions for Studying Argumentative Writing. In *The Proceedings of the Annual Meeting of the Association for Computational Linguistics*, pp. 1568-1578, Vancouver, Canada.
- (C.15) Changsheng Liu and Rebecca Hwa. 2017. Representations of Context in Recognizing the Figurative and Literal Usages of Idioms. In *The Proceedings of Thirty-First AAAI Conference on Artificial Intelligence*, San Francisco, California.
- (C.16) Changsheng Liu and Rebecca Hwa. 2016. Phrasal Substitution of Idiomatic Expressions. In *The Proceedings of The 15th Annual Conference of the NAACL/HLT*, San Diego, California.
- (C.17) Homa Hashemi and Rebecca Hwa, 2016. An Evaluation of Parser Robustness for Ungrammatical Sentences. In *The Proceedings of Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Austin, Texas.
- (C.18) Homa Hashemi and Rebecca Hwa. 2016. Parse Tree Fragmentation of Ungrammatical Sentences. In *The Proceedings of the 25th International Joint Conference on Artificial Intelligence (IJCAI)*, New York, New York.
- (C.19) Huichao Xue and Rebecca Hwa. 2014. Redundancy Detection in ESL Writings. In *The Proceedings of the 14th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, Gothenburg, Sweden.
- (C.20) Homa Hashemi and Rebecca Hwa. 2014. A Comparison of MT Errors and ESL Errors. In *the Proceedings of the 9th edition of the Language Resources and Evaluation Conference (LREC-2014)*, Reykjavik, Iceland.
- (C.21) Huichao Xue and Rebecca Hwa. 2012. Modeling ESL Word Choice Similarities By Representing Word Intentions and Extensions. In *The Proceedings of the 24th International Conference on Computational Linguistics (COLING-2012)*, Mumbai, India.
- (C.22) Matthias Grabmair, Kevin D. Ashley, Rebecca Hwa, Patricia M. Sweeney. 2011. Toward Extracting Information from Public Health Statutes using Text Classification Machine Learning. In *the Proceedings of the 24th International Conference on Legal Knowledge and Information Systems (Jurix-2011)*, Vienna, Austria.
- (C.23) Huichao Xue and Rebecca Hwa. 2010. Syntax-Driven Machine Translation as a Model of ESL Revision. In *The Proceedings of the 23th International Conference on Computational Linguistics (COLING-2010)*, Beijing, China.

- (C.24) Joshua S. Albrecht, Rebecca Hwa, and G. Elisabeta Marai. 2009. Correcting Automatic Translations through Collaborations between MT and Monolingual Target-Language Users. In *The Proceedings of the 12th Conference of the European Chapter of the ACL (EACL-2009)*, Athens, Greece.
- (C.25) Behrang Mohit, Frank Liberato, and Rebecca Hwa. 2009. Language Model Adaptation for Difficult-to-Translate Phrases. In *The Proceedings of the 13th Annual Conference of the European Association for Machine Translation (EAMT-09)*, Barcelona, Spain.
- (C.26) Joshua Albrecht and Rebecca Hwa. 2007. A Re-examination of Machine Learning Approaches for Sentence-Level MT Evaluation. In *The Proceedings of the 45th Annual Meeting of the Association for Computational Linguistics*, Prague, Czech Republic.
- (C.27) Joshua Albrecht and Rebecca Hwa. 2007. Regression for Sentence-Level MT Evaluation with Pseudo References. In *The Proceedings of the 45th Annual Meeting of the Association for Computational Linguistics*, Prague, Czech Republic.
- (C.28) Marcus Mylonakis, Khalil Sima'an, and Rebecca Hwa. 2007. Unsupervised Estimation for Noisy-Channel Models. In *The Proceedings of International Conference of Machine Learning (ICML)*, Corvallis, Oregon.
- (C.29) Rebecca Hwa, Carol Nichols, Khalil Sima'an. 2006. Corpus Variations for Translation Lexicon Induction. In *The Proceedings of the seventh conference of the Association for Machine Translation in the Americas*. Cambridge, MA.
- (C.30) Chenhai Xi and Rebecca Hwa. 2005. A Backoff Model for Bootstrapping Resources for Non-English Languages. In *The Proceedings of the Joint Conference of Human Language Technologies and Empirical Methods in Natural Language Processing (HLT/EMNLP-05)*, Vancouver, Canada.
- (C.31) Theresa Wilson, Janyce Wiebe, and Rebecca Hwa. 2004. Just how mad are you? Finding strong and weak opinion clauses. In *The Proceedings of The Nineteenth National Conference on Artificial Intelligence (AAAI-2004)*, San Jose, California.
- (C.32) Mark Steedman, Rebecca Hwa, Stephen Clark, Miles Osborne, Anoop Sarkar, Julia Hockenmaier, Paul Ruhlen, Steven Baker, and Jeremiah Crim. 2003. Example Selection for Bootstrapping Statistical Parsers. In *the Proceedings of the joint Conference of Human Language Technologies and the North American Chapter of the Association for Computational Linguistics (HLT/NAACL-2003)*, Edmonton, Canada.
- (C.33) Mark Steedman, Miles Osborne, Anoop Sarkar, Stephen Clark, Rebecca Hwa, Julia Hockenmaier, Paul Ruhlen, Steven. Baker, and Jeremiah Crim. 2003. Bootstrapping Statistical Parsers from Small Datasets. In *The Proceedings of the Tenth Conference of the European Chapter of the Association for Computational Linguistics*, Budapest, Hungary.
- (C.34) Rebecca Hwa, Philip Resnik, Amy Weinberg, and Okan Kolak. 2002. Evaluating Translational Correspondence using Annotation Projection. In *The Proceedings of the 40th Anniversary Meeting of the Association for Computational Linguistics*, Philadelphia, Pennsylvania.

- (C.35) Bonnie Dorr, Lisa Pearl, Rebecca Hwa, and Nizar Habash. 2002. DUSTer: A Method for Unraveling Cross-Language Divergences for Statistical Word-level Alignment. In *The Proceedings of the Fifth Conference of the Association for Machine Translation in the Americas (AMTA-2002)*, Tiburon, California.
- (C.36) Rebecca Hwa. 2001. On Minimizing Training Data for Parser Acquisition. In *The Proceedings of the Conference on Computational Natural Language Learning (CoNLL-2001)*.
- (C.37) Rebecca Hwa. 2000. Sample Selection for Grammar Induction. In *The Proceedings of the 2000 Joint SIGDAT Conference on Empirical Methods in Natural Language Processing and Very Large Corpora*, Hong Kong, China.
- (C.38) Rebecca Hwa. 1999. Supervised Grammar Induction using Training Data with Limited Constituent Information. In *The Proceedings of the 37th Annual Meeting of the Association for Computational Linguistics*, College Park, Maryland.
- (C.39) Rebecca Hwa. 1998. An Empirical Evaluation of Probabilistic Lexicalized Tree Insertion Grammars. In *the Proceedings of the 36th Annual Meeting of the Association for Computational Linguistics and 17th International Conference on Computational Linguistics (COLING-ACL 1998)*, volume 1, Montreal, Canada.

REFEREED SHORT PAPERS, WORKSHOP PAPERS, AND DEMO PAPERS

- (W.1) Diane Litman, Tazin Afrin, Omid Kashefi, Christopher Olshefski, Amanda Godley and Rebecca Hwa. 2022. Development and Classroom Evaluation of an Automated Writing Evaluation System for Supporting Self-Monitored Revising. In *The Proceedings of the 23rd International Conference on Artificial Intelligence in Education (AIED)*, Short Paper Track. July 2022, Durham, UK.
- (W.2) Omid Kashefi and Rebecca Hwa. 2021. Contrapositive Local Class Inference. In *The 7th Workshop on Noisy User-generated Text (W-NUT)*.
- (W.3) Mingda Zhang, Chun-Te Chu, Andrey Zhmoginov, Andrew Howard, Brendan Jou, Yukun Zhu, Li Zhang, Rebecca Hwa, Adriana Kovashka. 2021. BasisNet: Two-stage Model Synthesis for Efficient Inference. In *The 4th Workshop on Efficient Deep Learning for Computer Vision (ECV21), Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshop*, pages 3081-3090. [best paper]
- (W.4) Omid Kashefi and Rebecca Hwa. 2020. Quantifying the Evaluation of Heuristic Methods for Textual Data Augmentation. In *The 6th Workshop on Noisy User-generated Text (W-NUT)*.
- (W.5) Mingda Zhang, Keren Ye, Rebecca Hwa and Adriana Kovashka. 2020. Story Completion with Explicit Modeling of Commonsense Knowledge. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*.

- (W.6) Omid Kashefi, Andrew T. Lucas, and Rebecca Hwa. 2018. Semantic Pleonasm Detection. In *the Proceedings of The 16th Annual Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Short Paper Track)*, June 2018, New Orleans, LA.
- (W.7) Fan Zhang, Rebecca Hwa, Diane Litman, and Homa Hashemi. 2016. ArgRewrite: A Web-based Revision Assistant for Argumentative Writings. In *The Proceedings of NAACL-HLT (Demo)*, pp. 37-41, San Diego, California.
- (W.8) Gaurav Trivedi, Phuong Pham, Wendy Chapman, Rebecca Hwa, Janyce Wiebe, and Harry Hochheiser. 2015. Bridging the Natural Language Processing Gap: An Interactive Clinical Text Review Tool. Poster presented at the *AMIA Summit on Clinical Research Informatics (CRI 2015)*. San Francisco, California.
- (W.9) Huichao Xue and Rebecca Hwa. 2014. Improved Correction Detection in Revised ESL Sentences. In *The Proceedings of The 52nd Annual Meeting of the Association for Computational Linguistics (ACL short)*, 2014, Baltimore, Maryland.
- (W.10) Sisi Lu, Ye Ye, Rich Tsui, Xiaochu Liu, Rebecca Hwa. 2013. Feature Selection for 30-Day Heart Failure Readmission Prediction Using Clinical Drug Data. *NIPS Workshop on Machine Learning for Clinical Data Analysis and Healthcare*, Lake Tahoe, Nevada.
- (W.11) Phuong Pham, Janyce Wiebe, Rebecca Hwa, Wendy Chapman. 2013. Automated Annotation of Colonoscopy Reports. *AMIA Summit on Clinical Research Informatics (CRI 2013)*, San Francisco, California.
- (W.12) Alexander Conrad, Harry Hochheiser, Wendy Chapman, Rebecca Hwa, and Janyce Wiebe. 2013. Tightly-Coupled Natural Language Processing and Visualization for Electronic Medical Record Chart Review. *AMIA Summit on Clinical Research Informatics (CRI 2013)* San Francisco, California.
- (W.13) Alexander Conrad, Janyce Wiebe, and Rebecca Hwa. 2012. Recognizing Arguing Subjectivity and Argument Tags. In *the Proceedings of the Workshop on Extra-Propositional Aspects of Meaning in Computational Linguistics (ExProM '12)*, Jeju Island, Korea.
- (W.14) Frank Liberato, Behrang Mohit, and Rebecca Hwa. 2010. Improving Phrase-Based Translation with Prototypes of Short Phrases. In *The Proceedings of the 11th Annual Conference of the North American Chapter of the Association for Computational Linguistics*, Los Angeles.
- (W.15) Joshua S. Albrecht, Rebecca Hwa, and G. Elisabeta Marai. 2008. The Chinese Room : Understanding and Correcting Machine Translation. In *The Proceedings of IEEE Visualization 2008 Poster Compendium*.
- (W.16) Joshua Albrecht and Rebecca Hwa, 2008. The Role of Pseudo References in MT Evaluation. In *The Proceedings of the Third Workshop on Statistical Machine Translation*

- at the 46th Annual Meeting of the Association for Computational Linguistics, Columbus, Ohio.
- (W.17) Noah Smith, Michael Heilman, and Rebecca Hwa, 2008. Question Generation as a Competitive Undergraduate Course Project. In *The Proceedings of the NSF Workshop on the Question Generation Shared Task and Evaluation Challenge*, Arlington Virginia.
 - (W.18) Behrang Mohit and Rebecca Hwa. 2007. Localization of Difficult-to-Translate Phrases. In *The Proceedings of the 2nd Workshop on Statistical Machine Translations at the 45th Annual Meeting of the Association for Computational Linguistics*, Prague, Czech Republic.
 - (W.19) Behrang Mohit and Rebecca Hwa. 2005. Syntax-based Semi-Supervised Named Entity Tagging. In *The Proceedings of the 43rd Meeting of the Association for Computational Linguistics (Poster/Demo Session)*, Ann Arbor, Michigan.
 - (W.20) Carol Nichols and Rebecca Hwa. 2005. Word Alignment and Cross-Lingual Resource Acquisition. In *The Proceedings of the 43rd Meeting of the Association for Computational Linguistics (Poster/Demo Session)*, Ann Arbor, Michigan.
 - (W.21) Beatriz Maeireizo, Diane Litman, and Rebecca Hwa. 2004. Co-training for Predicting Emotions with Spoken Dialogue Data. In *The Proceedings of the 42nd Annual Meeting of the Association for Computational Linguistics Poster/Demo Session*. Boston, Massachusetts.
 - (W.22) Rebecca Hwa, Miles Osborne, Anoop Sarkar, and Mark Steedman. 2003. Corrected Co-training for Statistical Parsers. In *The Proceedings of the Workshop on the Continuum from Labeled to Unlabeled Data in Machine Learning and Data Mining at the International Conference of Machine Learning (ICML)*, Washington D.C..
 - (W.23) Rebecca Hwa, Philip Resnik, and Amy Weinberg. 2002. Breaking the Resource Bottleneck for Multilingual Parsing. In *The Proceedings of the Workshop on Linguistic Knowledge Acquisition and Representation: Bootstrapping Annotated Language Data, Third International Conference on Language Resources and Evaluation (LREC-2002)*, Las Palmas, Canary Islands, Spain.
 - (W.24) Adam Lopez, Michael Nossal, Rebecca Hwa, and Philip Resnik. 2002. Word-level Alignment for Multilingual Resource Acquisition. In *The Proceedings of the Workshop on Linguistic Knowledge Acquisition and Representation: Bootstrapping Annotated Language Data, Third International Conference on Language Resources and Evaluation (LREC-2002)*, Las Palmas, Canary Islands, Spain.

DISSERTATION

Rebecca Hwa. *Learning Probabilistic Lexicalized Grammars for Natural Language Processing*. Harvard University, 2001.

GRANTS AND CONTRACTS

- (G.1) LIFT-AI: Leadership and Innovation in Faculty Training for AI
Rebecca Hwa (PI), Adam Aviv (Co-PI), Shi Feng (Co-PI), Robert Pless (Co-PI), Arkady Yerukhimovich (Co-PI)
University of Hawaii (sponsor) / National Science Foundation (primary sponsor)
Amount awarded for the subcontract: \$699,799.00 for 24 months, starting February 2026.
- (G.2) PISCES 2023 - Partnership in Securing Cyberspace Through Education and Service
Arkady Yerukhimovich (PI), Adam Aviv (Co-PI), Rebecca Hwa* (Co-PI), Constantine Toregas (Co-PI)
National Science Foundation
Amount awarded: 5,391,208.00 for eight years, starting Sept. 2018
*Added as a Co-PI in 2026
- (G.3) Evacuate or Stay? Trustworthy LLMs for Disaster Decision Support
Erica Gralla (PI), Rebecca Hwa (Co-PI)
TRAILS Seed Grant (sponsor) / National Science Foundation (primary sponsor)
Amount awarded for the seed grant: \$148,228 for 12 months, starting Sept. 2025.
- (G.4) Characterizing and Countering the Normalization of Extremism and Communal Violence in Cyber-Social Space
Rebecca Hwa (PI)
University of Pittsburgh (sponsor) / Office of Naval Research (primary sponsor)
Amount awarded for the subcontract: \$12,214 for 18 months, starting Oct. 2023.
- (G.5) NRT-AI-FW-HTF: Co-Design of Trustworthy AI and Future Work Systems
Zoe Szajnfarder (PI), Rebecca Hwa* (Co-PI), Robert Pless (Co-PI), Ekundayo Shittu (Co-PI), Erica Wortham (Co-PI)
National Science Foundation
Amount awarded: \$3,000,000 for six years, starting Sept. 2021.
*Added as a Co-PI in 2024.
- (G.6) TriBal: A Tripartite Model for Group Bias Analytics
Yuru Lin (PI), Wen-Ting Chung (Co-PI), Rebecca Hwa (Co-PI)
Defense Advanced Research Project Agency
Amount awarded: \$912,072 for 18 months, starting June 2018.
- (G.7) Cyberlearning EXP: Development of Human Language Technologies to Improve Disciplinary Writing and Learning through Self-Regulated Revising
Rebecca Hwa* (PI), Amanda Godley (Co-PI), Diane Litman (Co-PI)
National Science Foundation
Amount awarded: \$549,555.00 for three years, starting August 2017.
* PI changed to Litman for 2019-2020 because Hwa is IPA at NSF.

- (G.8) RI Small: Modeling Vividness and Symbolism for Decoding Visual Rhetorics
Adriana Kovashka (PI), Rebecca Hwa* (Co-PI)
National Science Foundation
Amount awarded: \$449,978.00 for three years, starting September 2017.
* Not a Co-PI for 2019-2020 while Hwa was IPA at NSF.
- (G.9) EAGER: Computational Models of Essay Rewritings
Rebecca Hwa (PI), Diane Litman (Co-PI)
National Science Foundation
Amount awarded: \$299,913 for two years, starting July 2015.
- (G.10) R01: Interactive Search and Review of Clinical Records with Multi-layered Semantic Annotation
Wendy Chapman (PI), Rebecca Hwa (Co-PI), Janyce Wiebe (Co-PI)
National Institute of Health
Amount awarded for Pitt's subcontract: \$940,000 for four years, starting October 2011.
- (G.11) CAREER: Robust Parsing for New Domains and Languages
Rebecca Hwa (PI)
National Science Foundation
Amount awarded: \$499,984 for five years, starting July 2008.
- (G.12) RAVINE: Recombination, Aggregation, and Visualization of Information in News-worthy Expressions
Noah Smith (PI, CMU), Alan Black (Co-PI CMU), Frederick Crabbe (Co-PI USNA),
Rebecca Hwa (Co-PI)
Department of Interior
Amount awarded for Pitt's subcontract: \$48,689 for two years, starting April 2008.
- (G.13) NLP Foundational Studies and Ontologies for Syndromic Surveillance from ED Reports
Wendy Chapman (PI), Rebecca Hwa (Co-PI), Janyce Wiebe (Co-PI)
National Institute of Health
Amount awarded: Total of \$1,257,457 for three years, starting August 2007.
- (G.14) Collaborative: Discriminative Knowledge-Rich Language Modeling for Machine
Rebecca Hwa (Co-PI), Alon Lavie (Co-PI, CMU)
National Science Foundation
Amount awarded for Pitt: \$87,389 for three years, starting July 2007; and \$6,000 for REU Supplemental in 2008.
- (G.15) SGER: Learning Syntax-based Evaluation Metrics for Machine Translation
Rebecca Hwa (PI)
National Science Foundation
Amount awarded: \$194,344 for two years, starting July 2006; and \$11,113 for two REU Supplementals, in 2007.

PATENT

Apparatus for determining the structure of a hypermedia document using graph partitioning (with Joseph Marks and Stuart Shieber). U. S. patent number 5,546,517, August 1996.

MENTORING

PHD STUDENTS SUPERVISED

Alice Dragnea, on-going
Iyad Ait Hou, on-going
Meiqi Guo, PhD (Pitt) 2023 (now at Google)
Omid Kashefi, PhD (Pitt) 2022 (now at Meta)
Mingda Zhang, PhD (Pitt) 2022 (joint with Prof. Adriana Kovashka, now at Google)
Changsheng Liu, PhD (Pitt) 2019 (now at Meta)
Homa B. Hashemi, PhD (Pitt) 2017 (now at Microsoft)
Huichao Xue, PhD (Pitt) 2015 (now at Meta)
Behrang Mohit, PhD (Pitt) 2010 (now at Apple Siri-Search)

PHD DISSERTATION COMMITTEE MEMBERSHIP

Yejin Kim (GWU) 2026
Amal Alqahtani (GWU) 2026
Rana Salama (GWU) 2026
Muheng Yan (Pitt) 2026
Shabnam Behzad (Georgetown) 2024
Karin Cox (Pitt) 2024
John Sipple (GWU) 2023
Max Bender (Pitt) 2023
Tazin Afrin (Pitt) 2022
Christopher Thomas (Pitt) 2020
Nils Murrugarra Llerena (Pitt) 2019
Vineet Raghu (Pitt) 2019
Zahra Rahimi (Pitt) 2019
Gaurav Trivedi (Pitt) 2019
Charmgil Hong (Pitt) 2018
Huy Nguyen (Pitt) 2018
Yoonjung Choi (Pitt) 2017
Lingjia Deng (Pitt) 2017
Wencan Luo (Pitt) 2017
Fan Zhang (Pitt) 2017
Eric Heim (Pitt) 2016
Zitao Liu (Pitt) 2016
Michael Lipschultz (Pitt) 2015
Wenting Xiong (Pitt) 2015
Cem Akkaya (Pitt) 2014

Iyad Batal (Pitt) 2013
Kaihong Liu (Dept. of Biomedical Informatics, Pitt) 2012
Michal Valko (Pitt) 2011
Swapna Sundaran (Pitt) 2010
Tomas Singliar (Pitt) 2009
Mihai Rotaru (Pitt) 2009
Hua Ai (Pitt) 2009
Siqing Du (Pitt) 2008
Theresa Wilson (Pitt) 2008
Fazil Necip Ayan (University of Maryland) 2005

MASTER THESIS/PROJECT STUDENTS SUPERVISED

Iyad Ait Hou (GWU) 2024
Vivek Punjabi (Pitt) 2016
Sisi Lu (Pitt) 2015
Joshua Albrecht (Pitt) 2008
Carol Nichols (Pitt) 2006
Beatriz Maeireizo (joint with Prof. Diane Litman, Pitt) 2004

GRADUATE DIRECTED STUDIES SUPERVISED (OVER MULTIPLE SEMESTERS)

Karin Cox, (Pitt) 2017-2018 (joint with Prof. Bruce Childers)
Mahbahneh Eshaghzadeh Torbati, (Pitt) 2016-2018 (joint with Prof. Vanathi Gopalakrishnan)
Yuhuang Jiang, (Pitt) 2014, 2016-2017
Phuong Pham, (Pitt) 2012-2014 (joint with Prof. Janyce Wiebe)
Onur Cobanoglu, (Pitt) 2010
Frank Liberato, (Pitt) 2007-2010
Chenhai Xi, (Pitt) 2004-2009

UNDERGRADUATE RESEARCH SUPERVISED (OVER MULTIPLE SEMESTERS)

Daniel Zheng, (Pitt) 2018
Carter Burns, (USNA) 2018 (joint with Prof. Frederick L. Crabbe of USNA)
Reed Armstrong, (Pitt) 2016 (joint with Prof. Diane Litman)
Andrew Lucas, (Pitt) 2016
Nicolo Manfredi (Pitt) 2016 (joint with Prof. Diane Litman)
Ellen Gawryla, (Pitt) 2010
Jerry Maloney, (Pitt) 2009 (joint with Prof. Noah Smith of CMU)
Jarrett Billingsley, (Pitt) 2008
Joshua Albrecht, (Pitt) 2006-2007
Wesley Goodman, (Pitt) 2006
Carol Nichols, (Pitt) 2004-2005

SERVICE

ACADEMIC COMMITTEES

The George Washington University - Provost Search Committee (2025-2026)
The George Washington University - School of Engineering and Applied Science Executive Committee (2023-present)
The George Washington University - School of Engineering and Applied Science Outstanding Service Award Committee (2023)
The George Washington University - Department of Computer Science Faculty Search Committee (2023, 2024, 2025)
The George Washington University - Department of Computer Science Justice Equity Diversity and Inclusion Committee (2023-2025)
University of Pittsburgh Central Research Development Fund Reviewer (2018)
University of Pittsburgh - School of Arts and Sciences Goldwater Scholarship Nomination Committee (2017)
University of Pittsburgh - School of Arts and Sciences Ad Hoc Tenure Committee (2016)
University of Pittsburgh - School of Arts and Sciences Undergraduate Research Awards Selection Committee (2006 – 2008)
University of Pittsburgh - School of Arts and Sciences Honors College Advisory Board (2009-2011)
University of Pittsburgh - Department of Computer Science CS Day Committee (2013, 2016)
University of Pittsburgh - Department of Computer Science Faculty Search Committee (2010, 2015, 2018)
University of Pittsburgh - Department of Computer Science Graduate Admissions & Financial Aid Committee (2008-2010, 2013-2017)
University of Pittsburgh - Department of Computer Science Graduate Program & Examinations Committee (2003 – 2005, 2012 – 2014)
University of Pittsburgh - Department of Computer Science Undergraduate Advising Committee (2003 – 2008)

OUTREACH

National Center for Women & Information Technology Extension Services Client Cohort (2018)
Linguistic Society of America – Summer Institute Faculty (2011)
Technology Leadership Initiative – Lecture presentations (2006, 2008)
North American Computational Linguistics Olympiad – Tutorial Co-chair (2007)
Computer Research Association – Women Distributed Mentor Project (2004)

PROFESSIONAL ACTIVITIES

INVITED TALKS

“Symbolism and Meaning in Large Language Models”
The 38th North American Conference on Chinese Linguistics (NACCL-38), Washington DC, May 2026. [Keynote address]
“The Impact of Language Models on Processing Persuasive Media”

The 18th IEEE International Conference on Application of Information and Communication Technologies (AICT-2024), Turin, Italy, Sept. 2024. [Keynote address]
 “Computational Models of Persuasive Media”
 The George Washington University, Washington DC, October 2021.
 “Towards a Better Understanding of Processing Persuasive Texts”
 Department of Biomedical Informatics, University of Pittsburgh, PA September 2020.
 “Separating the Sheep from the Goats: On recognizing the Literal and Figurative Usages of Idioms”
 University of California Los Angeles, CA, March 2019.
 University of Southern California, CA, March 2019.
 Georgetown University, Washington DC, February 2018.
 University of Maryland at College Park, MD, November 2017.
 “Natural Language Processing and Its Applications to Healthcare”
 UPMC Enterprise, PA, September 2016.
 “Relationships between Machine Translation and Grammar Error Correction”
 IBM Watson, NY, March 2014.
 “The Role of Machine Translation in Modeling English as a Second Language (ESL) Writings”
 Universite de Montreal, Canada, February 2012.
 University of Maryland, MD, March 2012.
 “Applications of Information Visualization for Natural Language Processing,”
 Johns Hopkins University, MD, April 2011.
 University of Illinois at Urbana Champaign, IL, October 2010.
 Carnegie Mellon University, PA, May 2010.
 “Learning Evaluation Metrics for Sentence-Level Machine Translation”
 University of Texas, Austin, October 2008.
 Cornell University, NY, April 2008.
 Carnegie Mellon University, PA, September 2007.
 “Building Resources for Parsing Arabic Dialects”
 Carnegie Mellon University, PA, October 2005.
 “Semi-supervised Learning for Information Management”
 University of Maryland, College Park. MD, February 2005.
 “Breaking the Resource Bottleneck for Multilingual Processing,”
 University of Edinburgh, UK, IKG Summer School. September 2004.
 “A Continuum of Bootstrapping Methods for Statistical Parsing,”
 University of Rochester, NY, December 2003.
 “Semi-Supervised Approaches for Learning to Parse Natural Languages,”
 University of Michigan, MI, April 2003.
 Information Sciences Institute, CA, April 2003.
 University of Pittsburgh, PA, March 2003.
 “Sample Selection for Parser Induction,”
 University of Pennsylvania, Philadelphia, PA, January 2002.
 “Learning Probabilistic Lexicalized Grammars for Natural Language Processing,”
 Johns Hopkins University, Baltimore, MD, March 2001.
 Naval Research Laboratory, Washington D.C., July 2000.
 IBM Watson Research Laboratory, Yorktown Heights, NY, June 2000.
 University of Maryland, College Park, MD, June 2000.
 University of Chicago, IL, April 2000.

OFFICES

Association for Computational Linguistics SIGDAT (Special Interest Group on Linguistic Data & Corpus-based Approaches to Natural Language Processing) – Advisory Board (2018 – present)

North American Chapter of the Association for Computational Linguistics – Past Chair of the Executive Board (2012 – 2014)

North American Chapter of the Association for Computational Linguistics – Chair of the Executive Board (2010 – 2012)

North American Chapter of the Association for Computational Linguistics – Executive board member (2008 – 2010)

Computational Linguistics – Editorial board member (2003-2005)

CONFERENCE ORGANIZATION

The Conference on Empirical Methods in Natural Language Processing, Program Co-Chair (2017)

The Conference on Empirical Methods in Natural Language Processing, Tutorial Co-Chair (2016)

The Conference on Empirical Methods in Natural Language Processing, Parsing Track Area Co-Chair (2008)

Joint Conference of Human Language Technologies and the North American Chapter of the Association for Computational Linguistics, Machine Translation Track Area Chair (2007)

The 44th Annual Meeting of the ACL, Student Research Workshop, Faculty Advisor (2006)

The Joint Conference of Human Language Technologies and Empirical Methods in Natural Language Processing, Parsing Track Area Chair (2005)

SELECTED PROGRAMMING COMMITTEES

The Annual Meeting of the Association for Computational Linguistics, The Annual Meetings of the North American Chapter of the Association for Computational Linguistics, The Conferences of the European Chapter of the Association for Computational Linguistics, Conference on Empirical Methods in Natural Language Processing, International Conference on Computational Linguistics, Conference on Computational Natural Language Learning, The AAAI Conferences on Artificial Intelligence, International Joint Conference on Artificial Intelligence, Conference on Computer Vision and Pattern Recognition.

SELECTED JOURNAL REVIEWS

Transaction of the Association for Computational Linguistics, Journal of Artificial Intelligence, Journal of Natural Language Engineering, Computational Linguistics, Machine Learning.